

1:

$$A = \{1, 3, 12, 35\}$$

$$B = \{3, 7, 12, 20\}$$

$$C = \{x \mid x \text{ is a prime number}\}$$

a) $A \cap B = \{3, 12\}$

b) $(A \cup B) \setminus C = \{1, 12, 20, 35\}$

c) $A \cup (B \setminus C) = \{1, 3, 12, 20, 35\}$

'a' and 'b' are both subsets of 'c'. There are no disjoints between these three sets.

8) a)

$$(A \setminus B) \cap C = (A \wedge \neg B) \wedge C$$

$$(A \cap C) \setminus B = (A \wedge C) \wedge \neg B$$

$$(A \wedge \neg B) \wedge C = (A \wedge C) \wedge \neg B \quad \text{commutative}$$

b)

$$(A \cap B) \setminus B = (A \wedge B) \wedge \neg B$$

$$= A \wedge (B \wedge \neg B) \quad \text{commutative}$$

The resulting contradiction shows the set is empty, as nothing can satisfy the entry requirements.

c)

$$A \setminus (A \setminus B) = A \wedge \neg(A \wedge \neg B)$$

$$A \cap B = A \wedge B$$

$$A \wedge \neg(A \wedge \neg B)$$

$$= A \wedge (\neg A \vee B)$$

De Morgan

$$= (A \wedge \neg A) \vee (A \wedge B)$$

Distributive

$$= \text{False} \vee (A \wedge B)$$

Contradiction

$$= A \wedge B$$

Identity

$$A \wedge \neg(A \wedge \neg B) = A \wedge B$$

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$$\begin{aligned}B\Delta C &\rightarrow A\Delta(B\Delta C) \\ A\Delta B &\rightarrow (A\Delta B)\Delta C\end{aligned}$$

1.5: 2) a)

P = Mary gets a good apartment Price

A = Mary finds a nice apartment

M = Mary moves out

$$(P \wedge A) \rightarrow M$$

b)

C = good credit history

D = adequate down payment

M = you get a mortgage

$$M \rightarrow (C \wedge D)$$

c)

S = Someone stops Jhon

D = Jhon drops out

$$\neg S \rightarrow D$$

d)

A = x is dvsbl by 4

B = x is dvsbl by 6

P = x is prime

$$P \rightarrow \neg(A \vee B)$$

8) a)

The first phrase ' $(P \rightarrow Q) \wedge (Q \rightarrow R)$ ' implies the transitive that can be found in the second, $(P \rightarrow R)$. Because the transitive $(P \rightarrow R)$ doesn't involve 'Q' at all, the two biconditionals in the second phrase are needed to cover the cases in which the conditionals are false.

b)

For the statement to not be a tautology:

If 'P' is true, then 'Q' must be false to make the first conditional false. But if Q is false, then that means that $(Q \rightarrow R)$ is true by default.

If 'P' is false, then the first conditional is true by default, making the second irrelevant.

(a) $P \rightarrow (Q \rightarrow R)$

P	Q	R	$Q \rightarrow R$	$P \rightarrow (Q \rightarrow R)$
T	T	T	T	T
T	T	F	F	F
T	F	T	T	T
T	F	F	T	T
F	T	T	T	T
F	T	F	F	T
F	F	T	T	T
F	F	F	T	T

(b) $Q \rightarrow (P \rightarrow R)$

P	Q	R	$P \rightarrow R$	$Q \rightarrow (P \rightarrow R)$
T	T	T	T	T
T	T	F	F	F
T	F	T	T	T
T	F	F	F	T
F	T	T	T	T
F	T	F	T	T
F	F	T	T	T
F	F	F	T	T

(c) $(P \rightarrow Q) \wedge (P \rightarrow R)$

(d) $(P \wedge Q) \rightarrow R$

(e) $P \rightarrow (Q \wedge R)$

Comparing the outputs of the truth tables yields two distinct sets of logically equivalent formulas:

- Statements (a), (b), and (d) are logically equivalent.
- Statements (c) and (e) are logically equivalent.

P	Q	R	$P \rightarrow Q$	$P \rightarrow R$	$(P \rightarrow Q) \wedge (P \rightarrow R)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	F
T	F	F	F	F	F
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	T	T	T

P	Q	R	$P \wedge Q$	$(P \wedge Q) \rightarrow R$
T	T	T	T	T
T	T	F	T	F
T	F	T	F	T
T	F	F	F	T
F	T	T	F	T
F	T	F	F	T
F	F	T	F	T
F	F	F	F	T

P	Q	R	$Q \wedge R$	$P \rightarrow (Q \wedge R)$
T	T	T	T	T
T	T	F	F	F
T	F	T	F	F
T	F	F	F	F
F	T	T	T	T
F	T	F	F	T
F	F	T	F	T
F	F	F	F	T